CFA® 2026 Level I

Complete Study Guide

Alternative Investment, Portfolio Management & Ethics

This document is a condensed study guide. Please refer to the official CFA Institute curriculum for complete details.

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Part I

Alternative Investments

1 Alternative Investment Features, Methods, and Structures

1.1 Features and Categories of Alternative Investments

LOS 76.a: Describe features and categories of alternative investments.

What Are Alternative Investments? Alternative investments comprise various types of investments that do not fall under the heading of traditional investments (long-only investments in cash or publicly traded stocks and bonds).

Types of Alternative Investment Structures:

- Hedge funds
- Private equity funds
- Real estate investments
- Commodities
- Infrastructure
- Digital assets

Perceived Benefits:

- Risk reduction from diversification (low correlations with traditional investments)
- Possible higher returns from illiquid securities
- Potential exploitation of less efficient markets

Key Features Compared to Traditional Investments:

- More specialized knowledge required of investment managers
- Relatively low correlations with returns of traditional investments
- Less liquidity of assets held

- Longer time horizons for investors
- Larger size of investment commitments

Three Broad Categories:

Category	**Description**
Private Capital	Private equity (LBOs, venture capital) and private debt (direct lending, venture debt, distressed debt)
Real Assets	Real estate, infrastructure, natural resources (farmland, timberland, commodities), digital assets
Hedge Funds	Investment companies using leverage, long/short positions, derivatives; available only to qualified investors

Professor's Note: *Although correlations with traditional investments may be low on average, they may increase significantly during periods of economic stress.*

1.2 Investment Methods: Direct, Co-Investment, and Fund Investment

LOS 76.b: Compare direct investment, co-investment, and fund investment methods for alternative investments.

Fund Investing: Investing in a pool of assets alongside other investors using a fund manager who selects and manages investments using an agreed-upon strategy.

- Investors do not control asset selection or management
- Manager receives management fee + incentive fee
- Requires larger capital commitments for longer periods
- Less information on positions and returns
- Higher management fees

Co-Investing: Investor contributes to a pool of investment funds but also has the right to invest directly alongside the fund manager in some assets.

- Reduces overall fees while benefiting from manager's expertise
- Provides opportunity to gain skills for direct investing
- For managers: increases availability of funds and diversification

Direct Investing: Investor purchases assets independently rather than pooling funds or using an outside manager.

- **Advantages:** No fees to outside managers, more control over investment choices
- **Disadvantages:** Less diversification, higher minimum investments, greater expertise required

1.3 Investment Ownership and Compensation Structures

LOS 76.c: Describe investment ownership and compensation structures commonly used in alternative investments.

Limited Partnership Structure:

- **General Partner (GP):** Fund manager, makes all investment decisions, takes on liabilities
- **Limited Partners (LPs):** Investors, own partnership share proportional to investment, no liability beyond investment, no say in management
- Typically available only to **accredited investors**

Limited Partnership Agreement: Contains rules and operational details. **Side letters** contain special terms for specific LPs (e.g., excusal rights, most-favored-nation clauses).

Master Limited Partnership (MLP): Publicly traded limited partnership, most common in natural resources or real estate.

Fee Structures:

Fee Type	Description
Management Fee	1-2% of fund's assets. For private equity: percentage of committed capital (not invested capital). "Dry powder" = committed capital not yet drawn down.
Performance/Incentive Fee	Portion of profits ("carried interest"). Often includes hurdle rate (preferred return).
Hurdle Rate	Soft hurdle: fees on total increase. Hard hurdle: fees only on gains above hurdle rate.
Catch-Up Clause	GP "catches up" to receive specified percentage of first tranche of gains.
High-Water Mark	No performance fee on gains that only offset prior losses.

Waterfall	Deal-by-deal (American): profits distributed per deal. Whole-of-fund (European): LPs receive all until initial investment + hurdle returned.
Clawback Provision	LPs can recover excess incentive payments if later losses negate prior gains.

Example: Performance Fee Calculation:

- Fund return = 12%, hurdle rate = 8%, performance fee = 20%
- Soft hurdle: Fee = 20% × 12% = 2.4%
- Hard hurdle: Fee = 20% × (12% - 8%) = 0.8%

2 Alternative Investment Performance and Returns

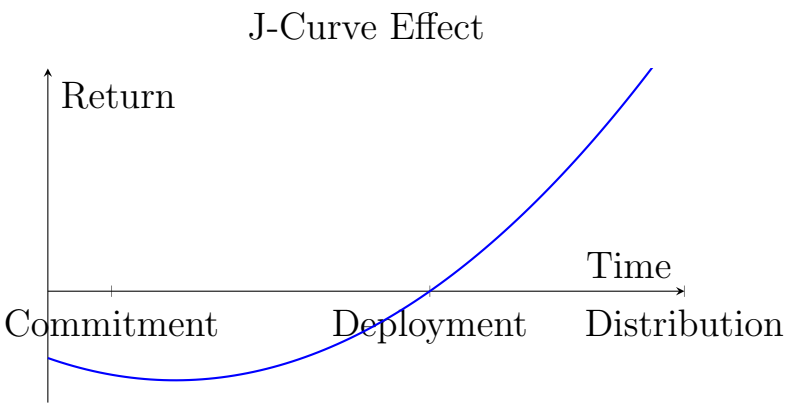
2.1 Performance Appraisal of Alternative Investments

LOS 77.a: Describe the performance appraisal of alternative investments.

Additional Risks in Alternative Investments:

- Timing of cash flows over investment life cycle
- Use of leverage by fund managers
- Valuation of illiquid investments (may not have observable market prices)
- Complexity of fees, taxes, and accounting

The J-Curve Effect:



Three Phases:

1. **Capital Commitment Phase:** Managers make capital calls, returns tend to be negative
2. **Capital Deployment Phase:** Managers fund and involve themselves in investments, returns remain negative
3. **Capital Distribution Phase:** Investments succeed, returns turn positive and accelerate

Performance Measures:

- **IRR (Money-Weighted Return):** Most appropriate because manager controls timing of cash flows
- **Multiple of Invested Capital (Money Multiple):** Ratio of total capital returned + remaining assets value to total capital paid in. Simpler but ignores timing

Leveraged Return Formula:

$$\boxed{\text{Leveraged Return} = \mathrm{r}(\mathrm{V}_0 + V_B) - r_B V_B \frac{}{V_0}}$$

Fair Value Hierarchy (Levels):

1. **Level 1:** Quoted prices in active markets (exchange-traded securities)
2. **Level 2:** Observable inputs (derivatives priced using models)
3. **Level 3:** Unobservable inputs (real estate, private equity)

Professor's Note: *Level 3 assets may have valuations that remain near initial cost for long periods, making reported returns appear higher, less risky, and less correlated than they really are.*

2.2 Return Calculations Before and After Fees

LOS 77.b: Calculate and interpret alternative investment returns both before and after fees.

Lockup Period and Notice Period:

- **Lockup period:** Time after initial investment when LPs cannot request redemptions (or incur significant fees)
- **Notice period:** Time fund has to fulfill redemption request (30-90 days)
- **Gate:** Discretionary restriction on redemptions

Simple Fee Calculation:

$$\boxed{\text{Total Fees} = \text{mV}_1 + \max[0, p(V_1 - V_0)]}$$

$$\boxed{\text{After-Fee Return} = \text{V}_1 - V_0 - \text{Total Fees}_{\overline{V_0}}}$$

Example: Hedge Fund Fees with High-Water Mark:

- Year 1: Beginning \$110M, management fee 2%, soft hurdle 5%, performance fee 20%
- Year 1 end value before fees = \$100.2M
- Management fee = \$2.2M, return after fee = -10.9% (no performance fee)
- Year 2 beginning = \$98M, end value before fees = \$119M
- Management fee = \$1.96M, value net of management fee = \$117.04M
- High-water mark = \$110M, gain above HWM = \$7.04M
- Performance fee = \$1.41M, total fees = \$3.37M
- After-fee return = 18.0%

Biases in Alternative Investment Returns:

- **Survivorship bias:** Failed funds not included → overstates returns, understates risk
- **Backfill bias:** Managers select successful funds for inclusion → overstates returns
- **Vintage year:** Year fund made first investment; compare funds with same vintage year

3 Investments in Private Capital: Equity and Debt

3.1 Private Equity Features and Characteristics

LOS 78.a: Explain features of private equity and its investment characteristics.

Private Equity Categories:

Category	Description
Leveraged Buyout (LBO)	Acquire public companies with large percentage of debt financing. "Go private" transactions.
Management Buyout (MBO)	Existing management team participates in purchase.
Management Buy-In (MBI)	Private equity manager replaces current management.
Venture Capital (VC)	Financing for early-stage companies. Gets common equity, convertible debt, or convertible preferred stock.
Formative Stage	Angel investing (idea stage), seed-stage (product development), early-stage (pre-production).
Later-Stage	Expansion venture capital after production/sales begin.
Mezzanine-Stage	Capital to prepare firm for IPO.
Minority Equity Investing	Less-than-controlling interest in public companies (PIPE - Private Investment in Public Equity).

Exit Strategies:

1. **Trade Sale:** Sell to strategic buyer. Faster, lower costs, but limited buyers.
2. **Public Listing:** IPO (most common), direct listing, or SPAC acquisition.
3. **Recapitalization:** Company issues debt to fund dividend distribution.

4. **Secondary Sale:** Sell to another private equity firm.
5. **Write-off/Liquidation:** Take losses from unsuccessful investment.

Risk and Return:

- Higher average returns than overall stock returns
- Higher standard deviation (more risk)
- Subject to survivorship and backfill bias
- Infrequent revaluation may bias volatility and correlation downward

3.2 Private Debt Features and Characteristics

LOS 78.b: Explain features of private debt and its investment characteristics.

Categories of Private Debt:

Category	Description
Direct Lending	Loans made directly to private companies. Senior and secured with covenants. Leveraged loans use borrowed money.
Venture Debt	Funding for start-up/early-stage firms. Often convertible or with warrants. Allows owners to maintain control.
Mezzanine Debt	Subordinated to senior secured debt. May have conversion rights or warrants.
Distressed Debt	Debt of mature companies in financial trouble (bankruptcy/default). Funds may restructure existing debt.
Unitranche Debt	Combines different classes of debt into single loan with blended interest rate. Ranks between senior and subordinated debt.

Key Characteristics:

- Higher returns than traditional bonds (compensates for higher risk and illiquidity)
- Low correlations with traditional investments (diversification benefit)
- Interest rate usually set relative to reference rate (e.g., SOFR)
- Higher risk includes illiquidity and default risk
- Requires specialized knowledge

3.3 Diversification Benefits of Private Capital

LOS 78.c: Describe the diversification benefits that private capital can provide.

Correlations with Public Markets: Correlations of private capital fund index returns with public market index returns range from 0.63 to 0.83.

Vintage Year:

- Year the fund made its first investment
- Performance greatly influenced by vintage year and business cycle phase
- Expansion vintage: Higher returns for early-stage companies
- Contraction vintage: Higher returns for distressed companies
- Investors should diversify across vintage years

Risk and Return Profiles (Highest to Lowest):

1. Private equity (highest risk/return)
2. Mezzanine debt
3. Unitranche debt
4. Senior direct lending
5. Senior real estate debt
6. Infrastructure debt (lowest)

4 Real Estate and Infrastructure

4.1 Real Estate Features and Characteristics

LOS 79.a: Explain features and characteristics of real estate.

Types of Real Estate:

- Single-family residential
- Commercial: office buildings, shopping centers, industrial/warehouse, rental residential

Real Estate Investment Quadrant:

- **Vertical dimension:** Public vs Private
- **Horizontal dimension:** Debt vs Equity

Direct Real Estate Investment: Benefits:

- Control over purchase, financing, improvements, marketing, sale
- Diversification (returns less than perfectly correlated with stocks/bonds)
- Tax benefits (depreciation deductions, interest expense deductions)

Drawbacks:

- Illiquidity and price opacity
- Complexity of managing property
- Need for specialized knowledge
- High initial investment
- Concentration risk

Indirect Real Estate Investment:

- Limited partnerships, joint ventures, publicly traded securities

- **REITs (Real Estate Investment Trusts):** Exempt from double taxation, exchange-traded
- Types: Equity REITs, Mortgage REITs, Hybrid REITs

REIT Strategies (Risk Increasing):

1. Core: High-quality commercial/residential, stable returns, open-end
2. Core-Plus: Modest development/redevelopment
3. Value-Add: Larger-scale development/redevelopment
4. Opportunistic: Large-scale redevelopment, distressed properties, speculation

4.2 Investment Characteristics of Real Estate

LOS 79.b: Explain the investment characteristics of real estate investments.

Risk and Return Characteristics:

- **Lowest risk:** First mortgages, investment-grade CMBSs (bond-like)
- **Next:** Core strategies (bond-like, stable rental income)
- **Highest risk:** Value-add and opportunistic strategies (equity-like)

REIT Benefits:

- Diversification across many properties
- Liquidity of publicly traded security
- Correlation with equity returns higher than direct investment
- Correlations increase during steep market downturns
- Improves risk-return profile of portfolio relative to traditional asset classes only

4.3 Infrastructure Features and Characteristics

LOS 79.c: Explain features and characteristics of infrastructure.

Types of Infrastructure Assets:

- Transportation: roads, airports, ports, railways
- Utility assets: gas distribution, electric generation/distribution, waste disposal
- Information/communication technology: telecom towers, cable systems
- Social infrastructure: prisons, schools, health care facilities

Cash Flow Types:

- Availability payments
- Usage-based payments (e.g., highway tolls)
- Take-or-pay arrangements

Investment Types:

- **Greenfield investments:** To be constructed. Follow build-operate-transfer (BOT) life cycle.
- **Brownfield investments:** Expand or privatize existing assets (e.g., sale-leaseback)
- **Secondary-stage investments:** Fully operational facilities, no further development needed

Investment Vehicles:

- Direct investment (low liquidity due to large scale)
- ETFs, mutual funds, private equity funds, MLPs
- Debt: private placement or publicly traded

4.4 Investment Characteristics of Infrastructure

LOS 79.d: Explain the investment characteristics of infrastructure investments.

Risk and Return by Investment Type:

Type	Characteristics
Brownfield (Secondary-stage)	Least risky, lowest return, stable cash flows, high current yields, little growth potential
Brownfield (Other)	More uncertainty, moderate risk/return
Greenfield	Most risky, greater growth potential, lower near-term yields
Demand-based projects	Riskiest (revenues depend on uncertain future demand)

Key Risks:

- Regulatory risk
- Financial leverage risk
- Construction risk (for greenfield)
- Operational risk (when privately operated)

Benefits:

- Stable cash flows from long-term contracts
- Low correlation with public equities
- Infrastructure debt safer and less affected by economic cycles
- Greenfield investments in developing economies have generated attractive long-term returns

Professor's Note: *Infrastructure investments are most suitable for long-term institutional investors (pension plans, life insurance companies, sovereign wealth funds).*

5 Natural Resources

5.1 Farmland, Timberland, and Raw Land

LOS 80.a: Explain features of raw land, timberland, and farmland and their investment characteristics.

Common Characteristics:

- All are illiquid investments
- Value primarily driven by location
- Proximity to transportation/markets increases value
- Access to water and soil quality matter for timberland/farmland
- All can generate lease income and price appreciation
- Farmland and timberland also generate income from output

Farmland:

- Primarily held by individuals (institutions invest in raw land and timberland)
- Fewer financing alternatives than residential/commercial real estate
- Newer REITs allow retail investors access
- Farmers use short positions in commodity futures to hedge expected harvest

Timberland:

- Lot sizes much larger than farmland
- Requires specialized expertise (TIMOs - Timberland Investment Management Organizations)
- Choice of when to harvest based on current prices and expected growth rates (flexibility option)

ESG Consideration: Agricultural crops (including trees) consume carbon → attractive to ESG-focused investors.

5.2 Commodities Features and Characteristics

LOS 80.b: Describe features of commodities and their investment characteristics.

Commodity Sectors:

- Metals (base/industrial, precious)
- Agricultural products
- Energy products

Investment Methods:

- Direct investment (physical commodities like grain, gold)
- Derivatives (futures, forwards, options on futures) - most common
- Exchange-Traded Products (ETFs, ETNs)
- Managed futures funds (CTAs - Commodity Trading Advisers)
- Separately Managed Accounts (SMAs) for larger investors
- Specialized funds in specific commodity sectors

Commodity Futures Pricing:

$$\boxed{\begin{gathered}\text{Futures Price} \approx \text{Spot Price} \times (1 + \text{Risk-Free Rate}) + \\ \text{Storage Costs} - \text{Convenience Yield}\end{gathered}}$$

Contango vs Backwardation:

- **Contango:** Futures price ¿ Spot price (low or no convenience yield). Decreases return for long-only investors.
- **Backwardation:** Futures price ¡ Spot price (high convenience yield). Increases return for long-only investors.

5.3 Sources of Risk, Return, and Diversification

LOS 80.c: Analyze sources of risk, return, and diversification among natural resource investments.

Commodity Price Drivers:

- **Demand:** Value to end users, global economic conditions
- **Supply:** Production, storage costs, existing inventories
- Supply is inelastic in short run (long lead times to alter production)
- Weather and plant disease affect agricultural commodities
- Extraction costs increase as more expensive methods/remote areas used

Historical Performance:

- Commodities: Higher returns and higher volatility than global stocks or bonds
- Timberland and farmland: Higher average returns, lower volatility than global stocks (similar to global bonds)

Diversification Benefits:

- Low correlations with global equities and bonds
- Commodity prices tend to move with inflation rates → hedge against inflation risk
- Commodities outperform stocks/bonds during high inflation, underperform during low inflation
- More sensitive to geopolitical and weather-related risk factors

6 Hedge Funds

6.1 Investment Features of Hedge Funds

LOS 81.a: Explain investment features of hedge funds and contrast them with other asset classes.

Key Features:

- Private pooled investment vehicles
- Generally available only to qualified/accredited investors
- Lightly regulated (unlike mutual funds)
- Managers have great freedom in selecting investment strategies
- Primary return drivers: market inefficiency and price volatility
- Invest in traditional asset classes, use leverage, use derivatives
- Often evaluated on total return or risk-adjusted return basis (not against benchmark)
- Managers often invest their own money
- Performance fees may be linked to high-water mark

Comparison with Other Vehicles:

	Hedge Funds	**Mutual Funds/REITs/ETFs**	**Private Equity**
Publicly traded	No	Yes	No
Regulation	Light	Heavy	Light
Liquidity	Low (lockup periods)	High	Very low
Time horizon	Shorter	Varies	Longer
Fees	High (2&20 typical)	Lower	High

Hedge Fund Strategy Categories:

1. **Equity Hedge Strategies:**
   - Fundamental long/short: Long undervalued, short overvalued (net long bias)
   - Fundamental growth: Long high-growth, short low-growth
   - Fundamental value: Long undervalued, short overvalued
   - Market neutral: Equal long/short, no market exposure
   - Short bias: Predominantly short positions (contrarian)
2. **Event-Driven Strategies:**
   - Merger arbitrage: Buy target, short acquirer
   - Distressed/restructuring: Buy securities of firms in financial distress
   - Activist shareholder: Buy sufficient shares to influence policies
   - Special situations: Invest in firms issuing/repurchasing securities, spinning off divisions, etc.
3. **Relative Value Strategies:**
   - Convertible arbitrage: Exploit pricing discrepancies between convertible bonds and common stock
   - Specific fixed income: ABS/MBS/high yield
   - General fixed income: Across issuers and types
   - Multistrategy: Across and within asset classes
4. **Opportunistic Strategies:**
   - Macro strategies: Based on global economic trends (equities, fixed income, currencies, commodities)
   - Managed futures: Commodity futures (CTAs) or financial futures

6.2 Investment Forms and Vehicles

LOS 81.b: Describe investment forms and vehicles used in hedge fund investments.

Commingled Funds (Master-Feeder Structure):

- Two feeder funds: offshore (tax haven) and onshore
- Both flow into master fund which makes investments
- Bypasses regional regulatory requirements
- Tax efficient, economies of scale

Separately Managed Account (SMA):

- Customized portfolio for single large investor
- Manager has no stake (interests less aligned)
- More operational oversight, appropriate for institutional investors
- Lower negotiated fees, but receives only most liquid trades

Fund-of-Funds (FoF):

- Investment company that invests in hedge funds
- **Advantages:** Diversification among strategies, manager expertise in selection, access to otherwise unavailable funds, reduced lockup periods, greater exit liquidity
- **Disadvantage:** Additional layer of fees (typically "1 and 10" on top of underlying fund fees)

6.3 Sources of Risk, Return, and Diversification

LOS 81.c: Analyze sources of risk, return, and diversification among hedge fund investments.

Three Sources of Hedge Fund Returns:

1. **Market Beta:** Return attributable to broad market index
2. **Strategy Beta:** Return attributable to specific sectors
3. **Alpha:** Additional return from manager skill through security selection

Use of Leverage: Hedge funds use leverage to magnify strategy beta and alpha. High fees act as drag on performance.

Biases in Hedge Fund Indexes:

- **Survivorship bias:** Poor-performing funds less likely to report; funds must exist minimum time/minimum size to be included
- **Selection bias:** Inconsistent category assignment or inclusion requirements
- **Backfill bias:** Adding prior years' returns when fund added to index (better historical returns more likely to be added)

Diversification Benefits:

- Performance highly dependent on time period measured
- Provide diversification benefits relative to traditional asset classes
- More correlated with equities than with fixed income

7 Introduction to Digital Assets

7.1 Distributed Ledger Technology (DLT)

LOS 82.a: Describe financial applications of distributed ledger technology.

Digital Assets: Assets that can be electronically created, stored, and transferred (cryptocurrencies, tokens, digital collectibles). Secured using DLT/blockchain technology.

Distributed Ledger Technology (DLT) Benefits:

- Accuracy, transparency, and security
- Rapid ownership transfer
- Peer-to-peer (P2P) interactions

DLT Disadvantages:

- Data protection concerns, potential privacy violations
- Large amount of computational power needed for verifying transactions

DLT Components:

- Digital ledger
- Consensus mechanism
- Network of participants

Smart Contracts: Computer programs that self-execute based on predetermined terms and conditions.

Consensus Protocols:

- **Proof of Work (PoW):** Miners solve cryptographic problems to verify transactions. Rewarded with cryptocurrency. Requires massive energy. Most widely used.

- **Proof of Stake (PoS):** Validators pledge collateral (staking) to guarantee validity. Rewarded with return on staked collateral.

Permissionless vs Permissioned Networks:

- **Permissionless (open):** Transactions visible to all, any user can execute transactions. No centralized authority.
- **Permissioned:** Users may be restricted from some activities. More cost-effective.

Types of Digital Assets:

- **Cryptocurrencies:** Digital currency issued privately, no central bank backing (Bitcoin, altcoins like Ether)
- **Stablecoins:** Linked to value of another asset (e.g., Tether pegged to USD). Algorithmic stablecoins control supply via algorithms.
- **Meme coins:** Launched for entertainment (Dogecoin)
- **CBDC (Central Bank Digital Currency):** Digital version of banknote issued by central bank
- **Tokens:** NFTs (nonfungible tokens), security tokens (ICOs), utility tokens, governance tokens

7.2 Investment Features of Digital Assets

LOS 82.b: Explain investment features of digital assets and contrast them with other asset classes.

Key Differences from Traditional Assets:

Feature	**Digital Assets vs Traditional**
Inherent Value	Most digital assets not backed by underlying assets or cash flows. No fundamental value. Prices determined by projected growth based on scarcity and ability to transfer value.
Transaction Validation	Traditional: private ledgers maintained by central intermediaries. Digital: decentralized digital ledgers.
Medium of Exchange	Traditional: fiat currencies. Digital: alternative to fiat, mainly for online transactions. Mainstream hesitant due to high costs, not legal tender, restrictions.
Regulation	Traditional: clear established rules. Digital: lacking well-developed standards.

7.3 Investment Forms and Vehicles

LOS 82.c: Describe investment forms and vehicles used in digital asset investments.

Direct Investment:

- Cryptocurrency wallet (digital encryption keys)
- Centralized exchanges (privately held, popular, but security vulnerabilities)
- Decentralized exchanges (distributed framework, harder to attack, difficult to regulate)

Risks of Direct Investment:

- Fraud (scam ICOs, pump-and-dump schemes, market manipulation, theft)
- Lost passkeys (estimated 20% of Bitcoin in lost/abandoned wallets)
- Whales (large entities) may manipulate prices

Indirect Investment:

- Cryptocurrency coin trusts (shares in trust holding cryptocurrency, similar to closed-end fund)
- Cryptocurrency futures contracts (cash settled, leverage)
- Cryptocurrency exchange-traded products (ETFs tracking digital assets)
- Cryptocurrency stocks (digital exchanges, payment providers, miners)
- Cryptocurrency hedge funds (discretionary long, long/short, quantitative, multistrategy)

Asset-Backed Tokens: Digital ownership of physical/financial assets. Can increase liquidity of high-priced assets via fractional ownership. Regulators generally classify as securities.

Decentralized Finance (DeFi): Marketplace of dApps (decentralized applications) offering medium for exchange, store of value, tokenization, immutable ownership records.

7.4 Risk, Return, and Diversification

LOS 82.d: Analyze sources of risk, return, and diversification among digital asset investments.

Key Drivers:

- Asset appreciation (limited supply, e.g., Bitcoin limited to 21 million units)
- Sometimes called "digital gold"
- High returns and high volatility compared to traditional asset classes

Historical Performance (Bitcoin vs Traditional):

- Average monthly return: Bitcoin 8.84%, S&P 500 1.13%
- Standard deviation: Bitcoin 0.32, S&P 500 0.04
- Coefficient of variation: Bitcoin 3.66, S&P 500 3.43

Diversification Benefits:

- Historically low correlations with traditional asset classes
- Correlations with S&P 500: Bitcoin 0.21
- Correlations may increase during periods of extreme market stress

Price Influences:

- Market adoption
- Risk appetite
- Technological innovation
- Network impacts
- Speculation
- Regulation

Part II

Portfolio Management

8 Portfolio Risk and Return: Part I

8.1 Historical Risk and Return of Major Asset Classes

LOS 83.a: Describe characteristics of the major asset classes that investors consider in forming portfolios.

Historical Risk and Return (U.S. 1926-2017):

Asset Class	Avg Annual Return (Geometric)	Standard Deviation
Small-cap stocks	12.1%	31.7%
Large-cap stocks	10.2%	19.8%
Long-term corporate bonds	6.1%	8.3%
Long-term government bonds	5.5%	9.9%
Treasury bills	3.4%	3.1%
Inflation	2.9%	4.0%

Key Conclusions:

- Asset classes with greatest average returns also have highest standard deviations (risk-return tradeoff)
- Returns do not follow normal distribution (negative skew, greater kurtosis/fatter tails)
- Negative skew: tendency towards large downside deviations
- Positive excess kurtosis: frequent extreme deviations on both sides
- Liquidity is an additional consideration (especially in emerging markets and infrequently traded securities)

8.2 Risk Aversion and Portfolio Selection

LOS 83.b: Explain risk aversion and its implications for portfolio selection.

Types of Investors:

- **Risk-Averse:** Dislikes risk. Given equal expected returns, chooses less risky investment. All financial models assume investors are risk-averse.
- **Risk-Seeking (Risk-Loving):** Prefers more risk. Given equal expected returns, chooses more risky investment.
- **Risk-Neutral:** Indifferent to risk. Given equal expected returns, indifferent between investments.

Example: Risk Aversion Example: Coin flip: heads get \$100, tails get \$0. Expected payoff = \$50.

- Risk-averse: Prefers certain \$50 over gamble
- Risk-seeking: Prefers gamble over certain \$50
- Risk-neutral: Indifferent between gamble and certain \$50

LOS 83.c: Explain the selection of an optimal portfolio, given an investor's utility (or risk aversion) and the capital allocation line.

Indifference Curves:

- Plot combinations of risk (standard deviation) and expected return among which investor is indifferent
- Slope upward for risk-averse investors (require higher expected return for higher risk)
- Steeper curves = more risk-averse
- Flatter curves = less risk-averse

Capital Allocation Line (CAL): Line representing combinations of risk-free assets and the optimal risky asset portfolio.

- Risk-free asset has zero standard deviation and zero correlation with risky portfolio
- Portfolio standard deviation $= w_A \times \sigma_A$
- Expected return $= w_A E(R_A) + (1 - w_A) R_f$

Two-Fund Separation Theorem: All investors' optimal portfolios will be made up of some combination of the optimal portfolio of risky assets and the risk-free asset.

Optimal Portfolio: Tangency point between indifference curve and CAL. Less risk-averse investors choose portfolios with more risk (higher allocation to risky assets).

8.3 Portfolio Standard Deviation

LOS 83.d: Calculate and interpret the mean, variance, and covariance (or correlation) of asset returns based on historical data.

Population Variance:

$$\sigma^2 = \frac{\sum_{t=1}^{T}(R_t - \mu)^2}{T}$$

Sample Variance:

$$s^2 = \frac{\sum_{t=1}^{T}(R_t - \bar{R})^2}{T-1}$$

Sample Covariance:

$$s_{XY} = \frac{\sum_{t=1}^{T}(R_{Xt} - \bar{R}_X)(R_{Yt} - \bar{R}_Y)}{T-1}$$

Correlation:

$$\rho_{XY} = \frac{s_{XY}}{s_X s_Y}, \quad -1 \leq \rho \leq +1$$

LOS 83.e: Calculate and interpret portfolio standard deviation.

Two-Asset Portfolio Variance:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

Two-Asset Portfolio Standard Deviation:

$$\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2}$$

LOS 83.f: Describe the effect on a portfolio's risk of investing in assets that are less than perfectly correlated.

Key Relationship:

- Greatest portfolio risk when $\rho = +1$ (perfect positive correlation)
- As correlation decreases from $+1$ to -1, portfolio risk decreases
- Lower correlation = greater diversification benefit
- If $\rho = -1$, portfolio risk can be eliminated completely for specific weights

LOS 83.g: Describe and interpret the minimum-variance and efficient frontiers of risky assets and the global minimum-variance portfolio.

Minimum-Variance Frontier: Set of portfolios that have the lowest standard deviation for each level of expected return.

Efficient Frontier: Portfolios with the greatest expected return for each level of risk. Coincides with top portion of minimum-variance frontier.

Global Minimum-Variance Portfolio: Portfolio on the efficient frontier with the least risk (leftmost point).

9 Portfolio Risk and Return: Part II

9.1 Capital Allocation Line and Capital Market Line

LOS 84.a: Describe the implications of combining a risk-free asset with a portfolio of risky assets.

Combining Risk-Free with Risky Portfolio:

- Risk of resulting portfolio = $w_{\text{risky}} \times \sigma_{\text{risky}}$
- Expected return = linear combination of risk-free rate and risky portfolio return
- Creates a straight line (CAL) from risk-free rate through risky portfolio

LOS 84.b: Explain the capital allocation line (CAL) and the capital market line (CML).

Capital Allocation Line (CAL): Line of possible portfolio risk/return combinations given risk-free rate and risk/return of a risky portfolio.

Capital Market Line (CML): Under homogeneous expectations, the optimal CAL for all investors. Uses market portfolio as the risky portfolio.

$$\boxed{\mathrm{E}(\mathrm{R}_P) = R_f + \left(\frac{E(R_M) - R_f}{\sigma_M}\right)\sigma_P}$$

Market Risk Premium:

$$\boxed{\mathrm{E}(\mathrm{R}_M) - R_f}$$

Borrowing and Lending:

- Lending portfolio: Left of market portfolio (invest in risk-free asset)
- Borrowing portfolio: Right of market portfolio (borrow at risk-free rate to invest more than 100% in market portfolio)

9.2 Systematic and Nonsystematic Risk

LOS 84.c: Explain systematic and nonsystematic risk, including why an investor should not expect to receive additional return for bearing nonsystematic risk.

Definitions:

- **Systematic Risk (Market Risk, Nondiversifiable Risk):** Risk that cannot be diversified away. Affects all securities (GDP growth, interest rates).
- **Unsystematic Risk (Unique Risk, Diversifiable Risk, Firm-Specific Risk):** Risk that can be eliminated by diversification. Specific to individual securities.

$$\boxed{\text{Total Risk} = \text{Systematic Risk} + \text{Unsystematic Risk}}$$

Why Unsystematic Risk Is Not Compensated:

- Diversification is essentially free (e.g., low-cost index funds)
- Investors will not be compensated for bearing risk that can be eliminated at no cost
- Equilibrium returns depend only on systematic risk

Diversification Graph: As number of stocks in portfolio increases, total risk decreases toward systematic risk level. Studies show 12-18 stocks achieve 90% of maximum diversification.

9.3 Return Generating Models and Beta

LOS 84.d: Explain return generating models (including the market model) and their uses.

Multifactor Model (General Form):

$$\boxed{\begin{gathered}\mathrm{E}(\mathrm{R}_i) - R_f = \\ \beta_{i1} \times E(\mathrm{Factor}_1) + \beta_{i2} \times E(\mathrm{Factor}_2) + \ldots + \beta_{ik} \times E(\mathrm{Factor}_k)\end{gathered}}$$

Factor Types:

- Macroeconomic (GDP growth, inflation, consumer confidence)
- Fundamental (earnings, earnings growth, firm size, research expenditures)
- Statistical (may have no theoretical basis, risk of data mining)

Fama-French Three-Factor Model:

1. Market excess return $(R_m - R_f)$
2. Firm size (SMB - Small Minus Big)
3. Book-to-market ratio (HML - High Minus Low)

Carhart Four-Factor Model: Adds momentum factor (prior period returns).

Market Model (Single-Index Model):

$$\boxed{\mathrm{R}_i = \alpha_i + \beta_i R_m + e_i}$$

LOS 84.e: Calculate and interpret beta.

Beta Formulas:

$$\boxed{\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\sigma_m^2} = \rho_{im}\left(\frac{\sigma_i}{\sigma_m}\right)}$$

Example: Beta Calculation:

- $\sigma_m = 20\%$, $\sigma_i = 30\%$, $\rho = 0.8$
- $\beta_i = 0.8 \times (0.30/0.20) = 1.2$

Interpretation:

- $\beta = 1$: Same systematic risk as market
- $\beta > 1$: More volatile than market
- $\beta < 1$: Less volatile than market
- $\beta = 0$: Uncorrelated with market

9.4 Capital Asset Pricing Model (CAPM) and Security Market Line (SML)

LOS 84.f: Explain the capital asset pricing model (CAPM), including its assumptions, and the security market line (SML).

Security Market Line (SML): Plots expected return versus systematic risk (beta or covariance).

CAPM Equation:

$$\boxed{\mathrm{E(R}_i) = R_f + \beta_i[E(R_m) - R_f]}$$

CAPM Assumptions:

1. Risk aversion (investors require higher return for higher risk)
2. Utility maximizing investors
3. Frictionless markets (no taxes, transaction costs, or impediments)
4. One-period horizon (all investors have same time horizon)
5. Homogeneous expectations (same estimates of risk, return, and correlations)
6. Divisible assets (infinitely divisible)
7. Competitive markets (no investor can influence prices)

CML vs SML:

	CML	**SML**
X-axis	Total risk (σ_p)	Systematic risk (beta)
What plots on it?	Only efficient portfolios	All properly priced securities and portfolios
Equation	$E(R_p) = R_f + \frac{E(R_m)-R_f}{\sigma_m}\sigma_p$	$E(R_i) = R_f + \beta_i[E(R_m) - R_f]$

9.5 Applications of CAPM and SML

LOS 84.g: Calculate and interpret the expected return of an asset using the CAPM.

Example: CAPM Calculation: $E(R_m) = 8\%$, $R_f = 2\%$, $\beta = 1.2$

$$E(R_A) = 2\% + 1.2(8\% - 2\%) = 9.2\%$$

LOS 84.h: Describe and demonstrate applications of the CAPM and the SML.

Identifying Mispriced Securities:

- Compare forecast return to CAPM required return
- Above SML → Undervalued → Buy
- Below SML → Overvalued → Short sell
- On SML → Properly valued

Example: Mispricing Example:

- $R_f = 7\%$, $E(R_m) = 15\%$
- Stock A: $\beta = 1.0$, forecast return = 12%
- Required return = 7% + 1.0(15%-7%) = 15%
- 12% ¡ 15% → Overvalued → Short sell

9.6 Performance Measures

LOS 84.i: Calculate and interpret the Sharpe ratio, Treynor ratio, M-squared, and Jensen's alpha.

Sharpe Ratio:

$$\boxed{\text{Sharpe Ratio} = \mathrm{R}_p - R_f \frac{}{\sigma_p}}$$

Measures excess return per unit of total risk.

Treynor Measure:

$$\boxed{\text{Treynor Measure} = \mathrm{R}_p - R_f \frac{}{\beta_p}}$$

Measures excess return per unit of systematic risk.

M-Squared (M^2):

$$\boxed{\mathrm{M}^2 = R_f + \frac{\sigma_m}{\sigma_p}(R_p - R_f)}$$

- Creates a leveraged/deleveraged portfolio with same risk as market
- M^2 alpha $= M^2 - R_m$
- Produces same rankings as Sharpe ratio (but in percentage terms)

Jensen's Alpha:

$$\boxed{\alpha_p = R_p - [R_f + \beta_p(R_m - R_f)]}$$

Measures excess return above SML for given beta.

Which Measure to Use?

- Single manager (unsystematic risk present): Use Sharpe ratio or M^2
- Multiple managers (well-diversified, no unsystematic risk): Use Treynor measure or Jensen's alpha

10 Portfolio Management: An Overview

10.1 Portfolio Approach and Management Process

LOS 85.a: Describe the portfolio approach to investing.

Portfolio Perspective: Evaluating individual investments by their contribution to the risk and return of an investor's portfolio (rather than in isolation).

Diversification Ratio:

$$\boxed{\text{Diversification Ratio} = \sigma_{\text{equally weighted portfolio}} \overline{\sigma_{\text{single security}}}}$$

Professor's Note: *Lower diversification ratio indicates greater risk-reduction benefit. Diversification works best when markets are normal; during turmoil, correlations increase, reducing benefits.*

LOS 85.b: Describe the steps in the portfolio management process.

Three Steps:

Step 1: Planning

- Analyze investor's risk tolerance, return objectives, time horizon, tax exposure, liquidity needs, income needs, unique circumstances
- Create Investment Policy Statement (IPS)
- Specify objective benchmark

Step 2: Execution

- Top-down analysis: Examine economic conditions, macroeconomic variables
- Asset allocation across asset classes
- Bottom-up security analysis: Identify undervalued securities

Step 3: Feedback

- Monitor changes in investor circumstances and asset characteristics
- Rebalance portfolio
- Measure and evaluate performance relative to benchmark

10.2 Types of Investors

LOS 85.c: Describe types of investors and distinctive characteristics and needs of each.

Investor	Risk Tolerance	Horizon	Liquidity Needs	Income Needs
Individuals	Depends	Depends	Depends	Depends
Banks	Low	Short	High	Pay interest
Endowments	High	Long	Low	Spending level
Insurance (Life)	Low	Long	Low	Pay claims
Insurance (P&C)	Low	Short	High	Pay claims
Mutual Funds	Depends	Depends	High	Depends
Defined Benefit Pension	High	Long	Low	Depends on age

LOS 85.d: Describe defined contribution and defined benefit pension plans.

Defined Contribution Plan:

- Employer contributes sum each period to employee's retirement account
- No promise about future value
- Employee makes investment decisions
- Employee assumes all investment risk

Defined Benefit Plan:

- Employer promises periodic payments after retirement
- Benefit based on years of service and final compensation

- Employer assumes investment risk
- Employer makes contributions to fund

10.3 Asset Management Industry and Pooled Investments

LOS 85.e: Describe aspects of the asset management industry.

Key Terminology:

- **Buy-side firms:** Manage investments for clients
- **Sell-side firms:** Broker-dealers, investment banks
- **Full-service asset managers:** Offer variety of investment styles and asset classes
- **Specialist asset managers:** Focus on particular style or asset class
- **Multi-boutique firm:** Holding company with multiple specialist managers

Active vs Passive Management:

- **Active management:** Attempts to outperform benchmark through manager skill
- **Passive management:** Attempts to replicate benchmark performance (index funds, smart beta)

Trends:

- Passive management market share growing (lower fees, questions about active manager value)
- Exponential growth in available data
- Robo-advisors (computer algorithms providing investment advice)

LOS 85.f: Describe mutual funds and compare them with other pooled investment products.

Mutual Funds:

- **Open-end fund:** Investors buy/sell shares at NAV. Newly invested cash is invested by managers.

- **Closed-end fund:** Fixed number of shares traded on exchanges. Market price can differ from NAV.
- **No-load funds:** No up-front or redemption fees
- **Load funds:** Charge up-front fees, redemption fees, or both

Types of Mutual Funds:

- Money market funds (short-term debt, low risk)
- Bond mutual funds (fixed-income securities)
- Stock mutual funds (index funds = passive, actively managed = higher fees, higher turnover)

Exchange-Traded Funds (ETFs):

- Similar to closed-end funds but traded on exchanges
- Most are passively managed (index tracking)
- Special redemption provisions keep market price close to NAV
- Can be sold short, purchased on margin, traded intraday
- Lower capital gains liability than open-end index funds

Separately Managed Account (SMA): Portfolio owned by single investor, managed according to that investor's needs. No shares issued.

Hedge Funds:

- Less regulated than mutual funds
- Limited number of investors
- Only sold to qualified investors (high minimum investments)

11 Basics of Portfolio Planning and Construction

11.1 Investment Policy Statement (IPS)

LOS 86.a: Describe the reasons for a written investment policy statement (IPS).

Purpose of IPS:

- Forces investor to articulate needs, circumstances, and constraints
- Ensures goals are realistic (risk and return must be compatible)
- Provides discipline and benchmark for evaluation

LOS 86.b: Describe the major components of an IPS.

IPS Components:

1. Description of client circumstances
2. Statement of Purpose
3. Statement of Duties and Responsibilities (manager, custodian, client)
4. Procedures for updates and responding to unforeseen events
5. Investment Objectives (return and risk)
6. Investment Constraints (time horizon, taxes, liquidity, legal/regulatory, unique needs)
7. Investment Guidelines (permitted asset types, leverage)
8. Evaluation and Review (benchmark, performance measurement)
9. Appendices (strategic asset allocation, rebalancing policy)

11.2 Risk and Return Objectives

LOS 86.c: Describe risk and return objectives and how they may be developed for a client.

Risk Objectives:

- **Absolute:** Specific limits (e.g., "no loss greater than 10% in any year")
- **Relative:** Relative to benchmark (e.g., "within 2% of FTSE return")
- Can be stated as probabilities (e.g., "95% probability of return ¿ -5%")

Return Objectives:

- **Absolute:** Specific percentage (e.g., "9% annual return")
- **Relative:** Outperform benchmark (e.g., "2% above S&P 500")
- Real returns (adjusted for inflation) vs nominal returns

LOS 86.d: Explain the difference between the willingness and the ability (capacity) to take risk.

Ability to Bear Risk: Based on financial circumstances (longer horizon, greater wealth, insurance, secure job → greater ability)

Willingness to Bear Risk: Based on investor's attitudes and beliefs (subjective, assessed via questionnaire)

Resolving Differences:

- If willingness > ability → Lower of the two prevails (ability)
- If ability > willingness → Educate investor; if unwilling, conform to lower (willingness)

11.3 Investment Constraints

LOS 86.e: Describe the investment constraints of liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances.

Constraints (R-R-T-T-L-L-U):

1. **Liquidity:** Need to turn assets into cash without price concessions
2. **Time Horizon:** Longer horizon → more risk and less liquidity acceptable
3. **Tax Situation:** Higher tax rates may prefer tax-free bonds or capital gains
4. **Legal and Regulatory:** Trust, corporate, qualified accounts may have restrictions
5. **Unique Circumstances:** Ethical preferences (responsible investing), religious preferences, diversification needs

Responsible Investing Approaches:

- Negative screening (exclude specific companies/industries)
- Positive screening (invest in companies with positive ESG practices)
- Thematic investing (select sectors to promote ESG goals)
- Impact investing (return + promote positive ESG practices)
- Engagement/active ownership (use share ownership to promote ESG)
- ESG integration (consider ESG throughout process)

11.4 Asset Allocation and Portfolio Construction

LOS 86.f: Explain the specification of asset classes in relation to asset allocation.

Asset Class Specification:

- Returns correlations within asset class should be relatively high
- Returns correlations between asset classes should be relatively low (diversification benefit)
- Examples: domestic/foreign equities, large/small cap, emerging/developed markets, bonds by maturity/credit rating

LOS 86.g: Describe the principles of portfolio construction and the role of asset allocation in relation to the IPS.

Strategic Asset Allocation: Baseline percentage allocations to asset classes designed to meet IPS objectives.

Tactical Asset Allocation: Deviations from strategic allocation to profit from short-term opportunities (active management).

Security Selection: Deviations from index weights within an asset class.

Risk Budgeting: Sets overall risk limit and allocates permitted risk to systematic risk, tactical asset allocation, and security selection.

Core-Satellite Approach:

- Core (majority): Passively managed indexes
- Satellite (smaller): Active strategies
- Reduces excessive trading and offsetting active positions

LOS 86.h: Describe how environmental, social, and governance (ESG) considerations may be integrated.

ESG Integration Considerations:

- Negative screening requires appropriate benchmark (exclude same companies)

- Positive screening/thematic/impact require customization
- For active ownership, clarify who votes shares and according to which factors
- ESG constraints may affect performance (both positive and negative effects possible)

12 The Behavioral Biases of Individuals

12.1 Cognitive Errors vs Emotional Biases

LOS 87.a: Compare and contrast cognitive errors and emotional biases.

Cognitive Errors: Due primarily to faulty reasoning or irrationality (statistical analysis errors, information processing errors, illogical reasoning, memory errors). Can possibly be reduced by increased awareness, training, or more information.

Emotional Biases: Stem from feelings, impulses, or intuition (not conscious thought). Difficult to overcome; may have to be accommodated.

Professor's Note: *A bias may have elements of both cognition and emotion. Focus on cognitive aspects for mitigation.*

12.2 Commonly Recognized Behavioral Biases

LOS 87.b: Discuss commonly recognized behavioral biases and their implications for financial decision making.

Cognitive Errors - Belief Perseverance:

Bias	Description
Conservatism	Maintain prior views, fail to update appropriately with new information
Confirmation	Seek information that supports prior beliefs, avoid conflicting information
Representativeness	Classify based on characteristics; includes base-rate neglect and sample-size neglect
Illusion of Control	Believe they can control outcomes when they cannot (often associated with overconfidence)
Hindsight	See past events as more predictable than they were ("I-knew-it-all-along")

Cognitive Errors - Information Processing:

Bias	Description
Anchoring	Base expectations on prior number, overweight its importance
Mental Accounting	View money in different accounts or from different sources differently
Framing	Decisions affected by how question or data is "framed"
Availability	Undue emphasis on information that is readily available or easily recalled

Emotional Biases:

Bias	Description
Loss Aversion	Feel more pain from loss than pleasure from equal gain

Overconfidence	Overestimate own intuitive ability (includes illusion of knowledge, self-attribution)
Self-Control	Lack discipline, favor short-term satisfaction over long-term goals (hyperbolic discounting)
Status Quo	Comfort with existing situation causes resistance to change
Endowment	Asset felt to be more valuable simply because it is already owned
Regret Aversion	Do nothing out of fear that actions could be wrong; herding behavior

12.3 Market Characteristics from Behavioral Biases

LOS 87.c: Describe how behavioral biases of investors can lead to market characteristics that may not be explained by traditional finance.

Bubbles and Crashes:

- Overconfidence → overtrading, underestimation of risk, lack of diversification
- Self-attribution + good results → fuels overconfidence
- Confirmation bias → ignore new information suggesting valuations won't continue to rise
- Anchoring → believe recent highs are rational even after prices decline
- Regret aversion → even skeptical investors stay in market

Value/Growth Anomaly:

- Value stocks historically outperformed growth stocks
- Fama-French: Extra returns compensation for additional risk (size and book-to-market)
- Behavioral explanation: Halo effect (representativeness) → overvaluation of growth stocks

Home Bias: Investors tend to invest heavily in domestic firms (less diversification). May result from belief in better information access or emotional desire.

13 Introduction to Risk Management

13.1 Risk Management Framework and Governance

LOS 88.a: Define risk management.

Risk Management Process:

1. Identify risk tolerance of organization
2. Identify and measure risks faced
3. Modify and monitor these risks

Key Insight: Risk management does not seek to minimize or eliminate all risks. Organization may increase exposure to risks it can manage well and decrease exposure to risks it cannot manage well.

LOS 88.b: Describe features of a risk management framework.

Framework Activities:

- Establishing processes and policies for risk governance
- Determining risk tolerance
- Identifying and measuring existing risks
- Managing and mitigating risks to achieve optimal bundle
- Monitoring risk exposures over time
- Communicating across the organization
- Performing strategic risk analysis

LOS 88.c: Define risk governance and describe elements of effective risk governance.

Risk Governance: Senior management's determination of risk tolerance, optimal risk exposure strategy, and framework for oversight of risk management function.

Risk Management Committee: Provides way for various parts of organization to bring up issues of risk measurement, integration, and mitigation.

13.2 Risk Tolerance and Risk Budgeting

LOS 88.d: Explain how risk tolerance affects risk management.

Factors Determining Risk Tolerance:

- Expertise in lines of business
- Skill at responding to negative outside events
- Regulatory environment
- Financial strength and ability to withstand losses

LOS 88.e: Describe risk budgeting and its role in risk governance.

Risk Budgeting: Process of allocating firm resources to assets/investments by considering their risk characteristics and how they combine to meet risk tolerance.

Risk Budget Approaches:

- Single metric (portfolio beta, VaR, duration, returns variance)
- Categories of investments (domestic equities, international debt, etc.)
- Specific risk factors (interest rate risk, equity market risk, FX risk)

13.3 Sources of Risk

LOS 88.f: Identify financial and non-financial sources of risk and describe how they may interact.

Financial Risks:

- **Credit risk:** Counterparty may not fulfill obligations
- **Liquidity risk:** Loss when selling at less than fair value
- **Market risk:** Uncertainty about market prices and interest rates

Non-Financial Risks:

- **Operational risk:** Human error, faulty processes, inadequate security, cyber risk
- **Solvency risk:** Organization runs out of cash
- **Regulatory risk:** Changes in regulatory environment
- **Political risk (tax risk):** Political actions imposing costs
- **Legal risk:** Uncertainty about exposure to legal action
- **Model risk:** Asset valuations based on incorrect models
- **Tail risk:** Extreme events more likely than analysis indicates
- **Accounting risk:** Accounting policies/estimates judged incorrect

Individual Risks:

- Mortality risk (life insurance)
- Longevity risk (lifetime annuity)
- Health care expenses (health insurance)

Interaction of Risks: Risks are not independent; they interact. Example: Market risk reduced with options → creates counterparty risk → may create legal risk → may create liquidity risk → may increase solvency risk.

13.4 Measuring and Modifying Risk Exposures

LOS 88.g: Describe methods for measuring and modifying risk exposures and factors to consider in choosing among the methods.

Risk Measures:

- **Standard deviation:** Volatility of asset prices (stand-alone risk)
- **Beta:** Market risk of equity securities (for well-diversified portfolios)
- **Duration:** Price sensitivity of debt securities to interest rates
- **Delta:** Sensitivity of derivatives to underlying asset price
- **Gamma:** Sensitivity of delta to underlying price
- **Vega:** Sensitivity to volatility of underlying price
- **Rho:** Sensitivity to risk-free rate

Tail Risk Measures:

- **Value at Risk (VaR):** Minimum loss over period with specific probability (e.g., 5% monthly VaR of $1M)
- **Conditional VaR (CVaR):** Expected loss given that loss exceeds minimum amount

Stress Testing and Scenario Analysis:

- Stress testing: Extreme change in single variable
- Scenario analysis: Changes in multiple inputs

Modifying Risk Exposures:

Method	Description
Avoid	Don't engage in activity with uncertain outcome
Prevent	Take actions to reduce or eliminate risk (e.g., data security)

Accept (Self-Insurance)	Bear risk; may establish reserve account
Diversify	More efficiently bear risk
Transfer (Insurance)	Another party takes on risk (insurance, surety bonds, fidelity bonds)
Shift (Derivatives)	Change distribution of outcomes (forwards, futures, options, swaps)

Professor's Note: *Organizations may use multiple methods for a single risk. Choose based on cost-benefit comparison.*

Part III

Ethical and Professional Standards

14 Ethics and Trust in the Investment Profession

14.1 Ethics and Professionalism

LOS 89.a: Explain ethics.

Ethics: Set of shared beliefs about what is good/acceptable behavior and what is bad/unacceptable behavior. Conduct that follows moral principles and improves outcomes for stakeholders.

LOS 89.b: Describe the role of a code of ethics in defining a profession.

Code of Ethics: Written set of moral principles that guide behavior. Communicates values, principles, and expectations. Provides general guide to acceptable behavior.

LOS 89.c: Describe professions and how they establish trust.

Profession Characteristics:

- Code and standards for professional behavior
- Regulatory body to enforce rules
- Focus on client needs
- Focus on service to society
- Requirement to put client interests first
- Continuing education requirement
- High standards of expertise and skill
- Monitoring of professional conduct

LOS 89.d: Describe the need for high ethical standards in investment management.

Why Ethics Matter in Investment Management:

- Investment management is an intangible product (difficult to evaluate)

- Unethical behavior can significantly impact client financial well-being
- Loss of trust reduces investment and increases required returns
- Misallocation of capital reduces economic growth

LOS 89.e: Explain professionalism in investment management.

Suitability vs Fiduciary Standard:

- **Suitability:** Match between client requirements and security characteristics
- **Fiduciary:** Use knowledge and expertise to act in client's best interests (stronger standard)

14.2 Challenges and Framework

LOS 89.f: Identify challenges to ethical behavior.

Challenges:

- Overrate ethical quality of own behavior
- Overemphasize personal traits vs situational influences
- Social pressure from others
- Loyalty to employer/coworkers
- Prospect of more money or prestige
- Rules-based compliance culture (ask "what can we do?" not "what should we do?")

LOS 89.g: Compare and contrast ethical standards with legal standards.

Key Differences:

- Not all unethical actions are illegal
- Not all illegal actions are unethical (civil disobedience, whistle-blowing)
- Ethical principles often set higher standard than laws
- New laws often result from perceived unethical behavior
- Ethical decisions require more judgment

LOS 89.h: Describe a framework for ethical decision making.

Ethical Decision-Making Framework:

1. **Identify:** Relevant facts, stakeholders, duties owed, ethical principles, conflicts of interest
2. **Consider:** Situational influences, additional guidance, alternative actions
3. **Decide and act**
4. **Reflect:** Was outcome as anticipated? Why or why not?

15 Code of Ethics and Standards of Professional Conduct

LOS 90.a: Describe the structure of the CFA Institute Professional Conduct Program and the process for enforcement.

Professional Conduct Program:

- Based on fairness of process and confidentiality
- Board of Governors has overall responsibility
- Disciplinary Review Committee enforces Code and Standards

Circumstances Prompting Inquiry:

1. Self-disclosure on Professional Conduct Statements
2. Written complaints
3. Evidence from public sources (media)
4. Report by exam proctor
5. Analysis of exam materials and social media

Possible Outcomes:

- No disciplinary sanctions
- Cautionary letter
- Discipline (member may accept or reject; rejection leads to hearing)

LOS 90.b: Identify the six components of the Code of Ethics and the seven Standards of Professional Conduct.

Code of Ethics (Six Components): Members and Candidates must:

1. Act with integrity, competence, diligence, respect, and ethically
2. Place integrity of profession and client interests above personal interests

3. Use reasonable care and exercise independent professional judgment
4. Practice and encourage others to practice professionally and ethically
5. Promote integrity and viability of global capital markets
6. Maintain and improve professional competence

Standards of Professional Conduct (Seven):

1. I. Professionalism
2. II. Integrity of Capital Markets
3. III. Duties to Clients
4. IV. Duties to Employers
5. V. Investment Analysis, Recommendations, and Actions
6. VI. Conflicts of Interest
7. VII. Responsibilities as a CFA Institute Member or CFA Candidate

LOS 90.c: Explain the ethical responsibilities required by the Code and Standards, including the subsections of each Standard.

Standard I - Professionalism:

- I(A) Knowledge of the Law: Comply with all applicable laws, rules, regulations. Most strict prevails.
- I(B) Independence and Objectivity: Maintain independence; don't accept gifts that could compromise objectivity.
- I(C) Misrepresentation: Don't knowingly make misrepresentations.
- I(D) Misconduct: Don't engage in dishonesty, fraud, deceit, or acts reflecting adversely.
- I(E) Competence: Maintain competence for professional responsibilities.

Standard II - Integrity of Capital Markets:

- II(A) Material Nonpublic Information: Don't act or cause others to act on MNPI.
- II(B) Market Manipulation: Don't engage in practices that distort prices or volume with intent to mislead.

Standard III - Duties to Clients:

- III(A) Loyalty, Prudence, and Care: Act for benefit of clients; place client interests first.
- III(B) Fair Dealing: Deal fairly and objectively with all clients.
- III(C) Suitability: Make reasonable inquiry into client's situation; determine suitability before recommending.
- III(D) Performance Presentation: Ensure performance information is fair, accurate, complete.
- III(E) Preservation of Confidentiality: Keep client information confidential (exceptions: illegal activities, required by law, client permits).

Standard IV - Duties to Employers:

- IV(A) Loyalty: Act for benefit of employer; don't deprive employer of advantage of skills.
- IV(B) Additional Compensation Arrangements: Disclose and get written consent for outside compensation.
- IV(C) Responsibilities of Supervisors: Make reasonable efforts to ensure supervisees comply.

Standard V - Investment Analysis, Recommendations, and Actions:

- V(A) Diligence and Reasonable Basis: Exercise diligence, have reasonable basis for recommendations.

- V(B) Communication with Clients: Disclose services, costs, investment processes, risks, limitations.
- V(C) Record Retention: Maintain appropriate records to support analysis and recommendations.

Standard VI - Conflicts of Interest:

- VI(A) Disclosure of Conflicts: Avoid or fully disclose conflicts.
- VI(B) Priority of Transactions: Client and employer transactions have priority over personal transactions.
- VI(C) Referral Fees: Disclose any compensation for referrals.

Standard VII - Responsibilities as a CFA Institute Member or Candidate:

- VII(A) Conduct as Participants in CFA Programs: Don't compromise reputation or integrity of CFA programs.
- VII(B) Reference to CFA Institute, CFA Designation, and CFA Program: Don't misrepresent or exaggerate meaning of CFA designation.

16 Guidance for Standards I-VII - Summary

Standard I: Professionalism - Key Violations

Subsection	Key Violations
I(A) Knowledge of Law	Violating any applicable law; failing to dissociate from violations
I(B) Independence & Objectivity	Accepting gifts that compromise objectivity; issuer-paid research without disclosure
I(C) Misrepresentation	Plagiarism; guaranteeing returns; misleading performance presentation
I(D) Misconduct	Dishonesty, fraud, deceit; acts reflecting adversely on professional reputation
I(E) Competence	Accepting role without necessary competence

Standard II: Integrity of Capital Markets

II(A) Material Non-public Info	Trading or causing others to trade on MNPI
II(B) Market Manipulation	Transactions intended to mislead market participants

Standard III: Duties to Clients

III(A) Loyalty, Prudence, Care	Not acting in client's best interest; soft dollars not benefiting client
III(B) Fair Dealing	Discriminating among clients in disseminating recommendations
III(C) Suitability	Recommending unsuitable investments; not gathering client information
III(D) Performance Presentation	Misstating performance; using non-comparable benchmarks
III(E) Preservation of Confidentiality	Disclosing client information without permission or legal requirement

Standard IV: Duties to Employers

IV(A) Loyalty	Soliciting employer's clients before leaving; misappropriating trade secrets
IV(B) Additional Compensation	Accepting outside compensation without written employer consent
IV(C) Supervisors	Failing to detect or prevent violations by supervisees

Standard V: Investment Analysis, Recommendations, Actions

V(A) Diligence & Reasonable Basis	Lack of adequate research; using third-party research without review
V(B) Communication	Failing to disclose risks, limitations, or investment process changes
V(C) Record Retention	Failing to maintain supporting records (7 years recommended)

Standard VI: Conflicts of Interest

VI(A) Disclosure of Conflicts	Failing to disclose ownership, compensation arrangements, board service
VI(B) Priority of Transactions	Personal trading before client trades (front-running)
VI(C) Referral Fees	Failing to disclose referral compensation

Standard VII: Responsibilities as CFA Member/Candidate

VII(A) Conduct in CFA Programs	Cheating; revealing exam content; violating exam policies
VII(B) Reference to CFA Designation	Implying superior performance; misusing designation after lapsed membership

17 Global Investment Performance Standards (GIPS)

LOS 92.a: Explain why GIPS were created, who can claim compliance, and who benefits.

Problems GIPS Address:

- Choosing top-performing portfolio to represent firm results
- Excluding terminated accounts (likely poor performers)
- Selecting favorable time periods

Who Can Claim Compliance: Investment management firms only (managing assets for others). Must comply fully on firmwide basis.

LOS 92.b: Describe the key concepts of the GIPS Standards for Firms.

Eight Sections:

1. Fundamentals of Compliance
2. Input Data and Calculation Methodology
3. Composite and Pooled Fund Maintenance
4. Composite Time-Weighted Return Report
5. Composite Money-Weighted Return Report
6. Pooled Fund Time-Weighted Return Report
7. Pooled Fund Money-Weighted Return Report
8. GIPS Advertising Guidelines

LOS 92.c: Explain the purpose of composites in performance reporting.

Composite: Grouping of discretionary portfolios representing similar investment strategy, objective, or mandate.

- Must include all fee-paying, discretionary portfolios (current and past)
- Portfolio assigned to composite before performance known (ex ante)
- Composite returns are asset-weighted (not simple average)

LOS 92.d: Describe fundamentals of compliance, including definition of firm and discretion.

Firm Definition: Corporation, subsidiary, or division held out to clients as business entity. Must include all geographic locations.

Discretion Definition: Firm determines which portfolios are discretionary/nondiscretionary. Nondiscretionary if client restrictions prevent carrying out intended strategy.

LOS 92.e: Describe the concept of independent verification.

Verification:

- Encouraged but not required
- Must be performed by third party (not firm itself)
- Applies to entire firm, not selected composite
- Verifier attests to firmwide compliance and proper processes/procedures

18 Ethics Application - Key Cases

LOS 93.a, 93.b: Evaluate practices, policies, and conduct relative to the Code and Standards.

Key Case Examples:

Standard	**Violation Example**
I(A) Knowledge of Law	Failing to investigate potential money-laundering in client account
I(B) Independence	Contributing to politician's campaign for preferential treatment
I(C) Misrepresentation	Guaranteeing specific returns; misleading social media posts
I(D) Misconduct	Using error-correction policy to make own performance look better
II(A) MNPI	Overhearing phone conversation about takeover; selective disclosure to analysts
II(B) Market Manipulation	Including fake shareholders to meet listing requirements
III(A) Loyalty	"Opting out" of best interest requirement in client agreements
III(C) Suitability	Making client-requested trade without investigating suitability
III(E) Confidentiality	Downloading client data to personal server (got hacked)
IV(A) Loyalty	Taking client list when leaving firm (property of employer)
IV(C) Supervision	Accepting CCO role without authority to enforce policies
V(A) Diligence	Using another's research without independent analysis
V(B) Communication	Changing CMBS rating methodology without disclosure

VI(A) Conflicts	Receiving payments from subadvisors without disclosure
VI(B) Priority	Front-running client trades; allocating profitable trades to personal accounts
VII(A) CFA Programs	Revealing which topics were/were not tested on exam
VII(B) CFA Designation	Using CFA designation after lapsed membership

Annexure: Numerical Graphs

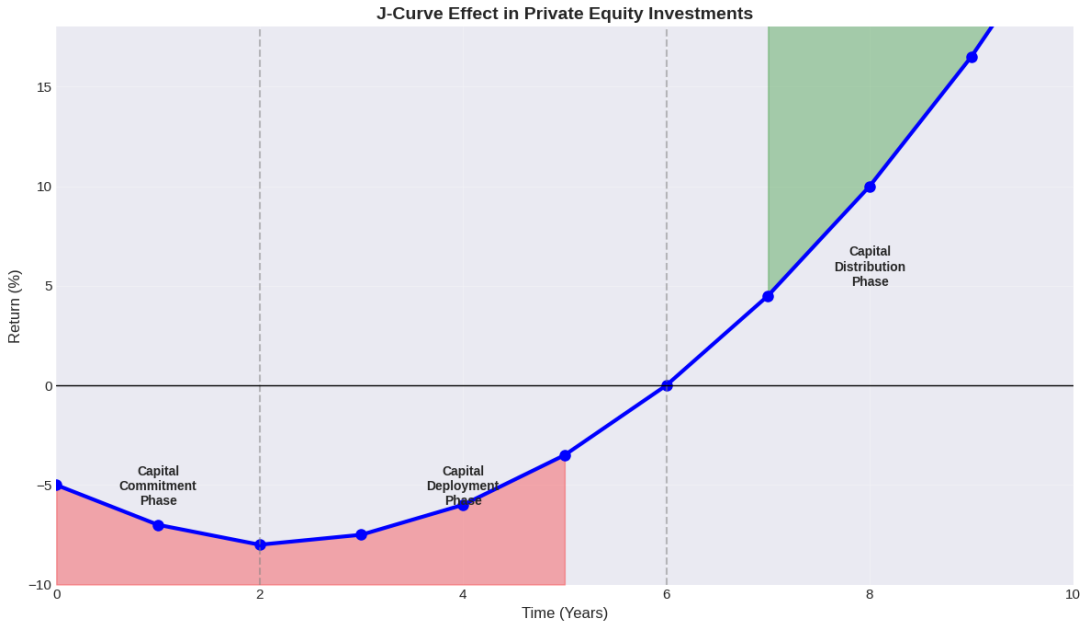


Figure 1: J-Curve Effect in Private Equity Investments - Shows the typical return pattern over time with initial negative returns followed by positive acceleration

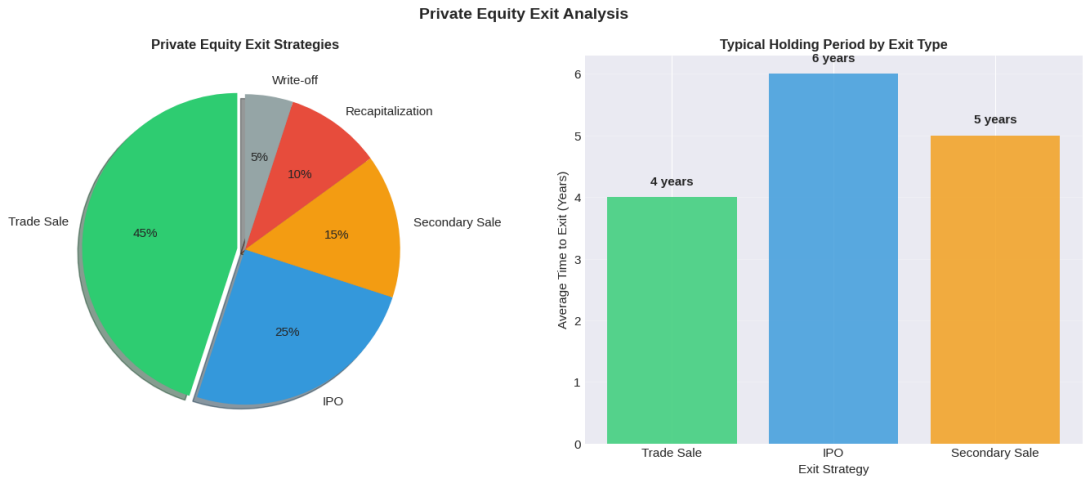


Figure 2: Private Equity Exit Analysis - Distribution of exit strategies and typical holding periods by exit type

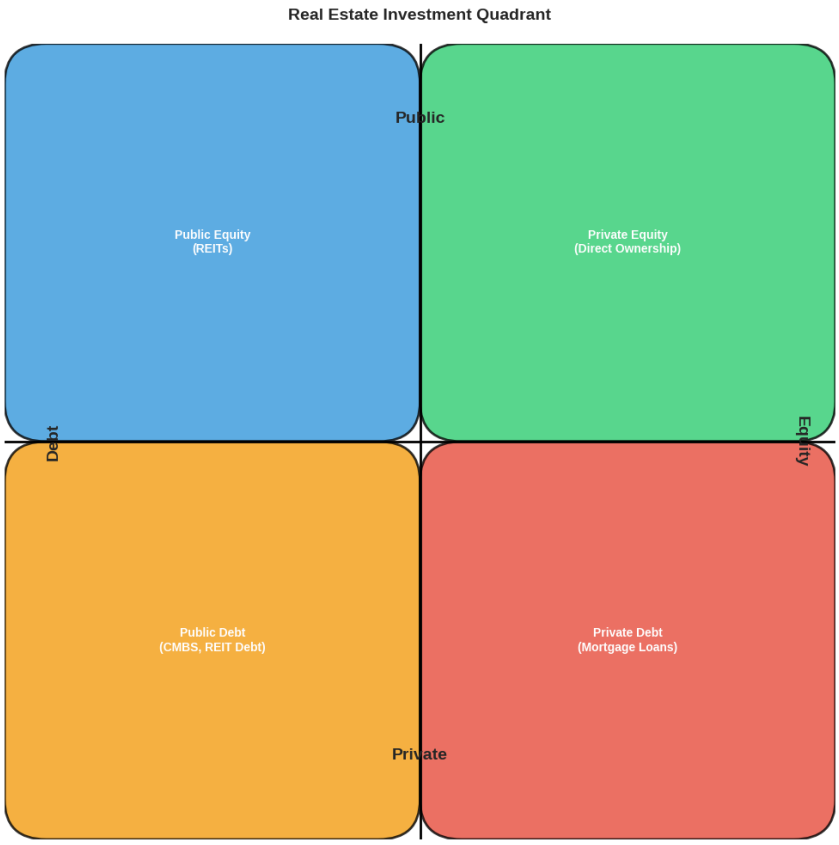


Figure 3: Real Estate Investment Quadrant - Categorizing real estate investments by Public/Private and Debt/Equity dimensions

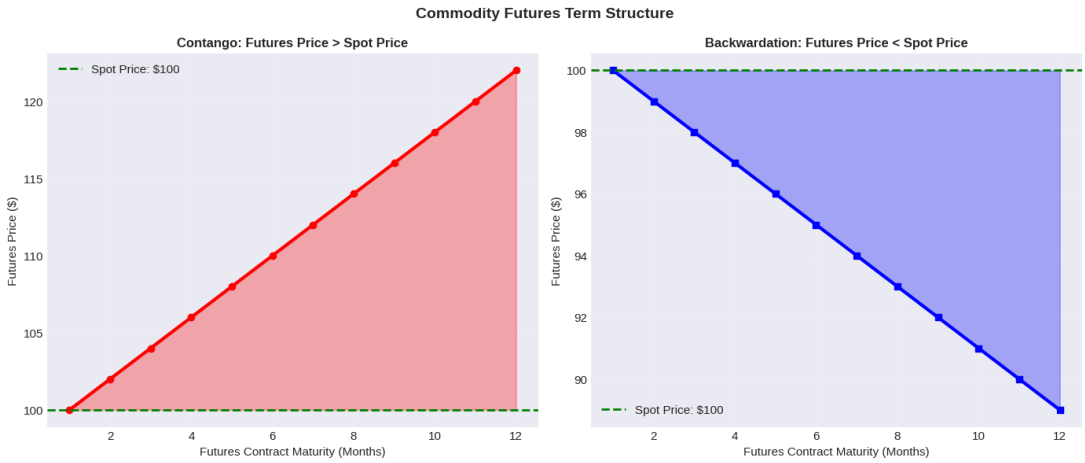


Figure 4: Commodity Futures Term Structure - Contango vs Backwardation price curves

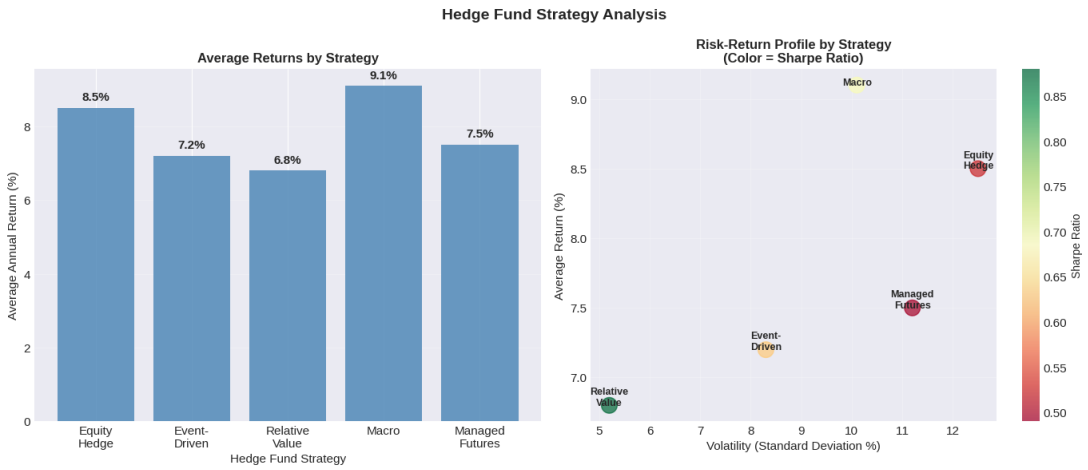


Figure 5: Hedge Fund Strategy Analysis - Average returns and risk-return profiles by strategy type

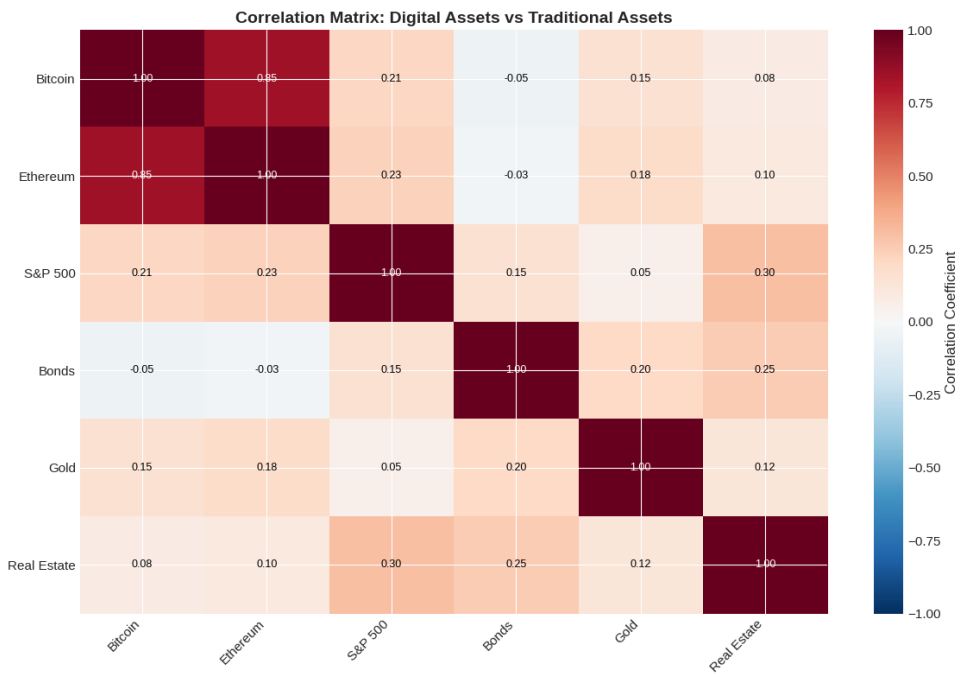


Figure 6: Correlation Matrix: Digital Assets vs Traditional Assets

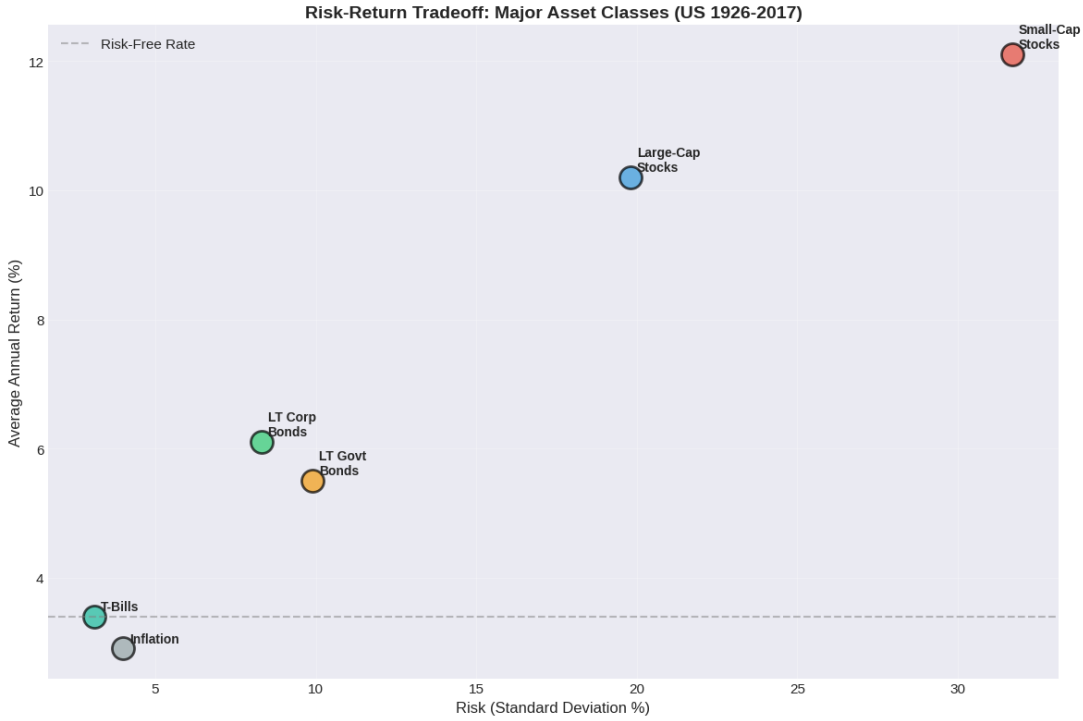


Figure 7: Risk-Return Tradeoff: Major Asset Classes (US 1926-2017) - Historical annual returns and standard deviations

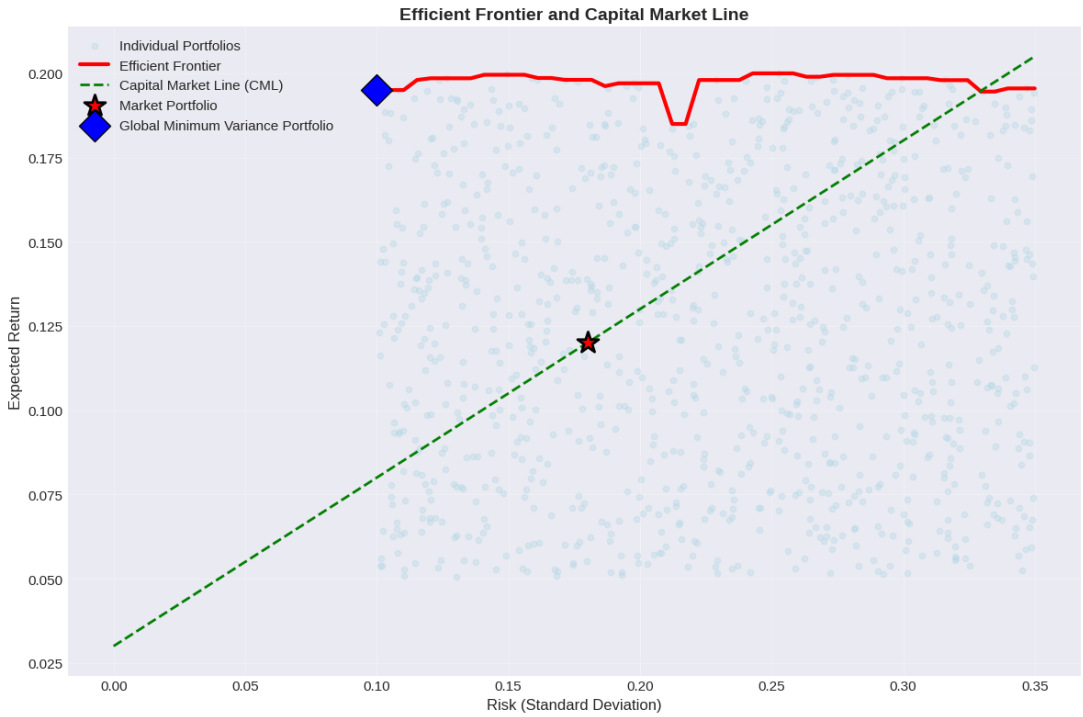


Figure 8: Efficient Frontier and Capital Market Line - Modern Portfolio Theory visualization showing optimal portfolios

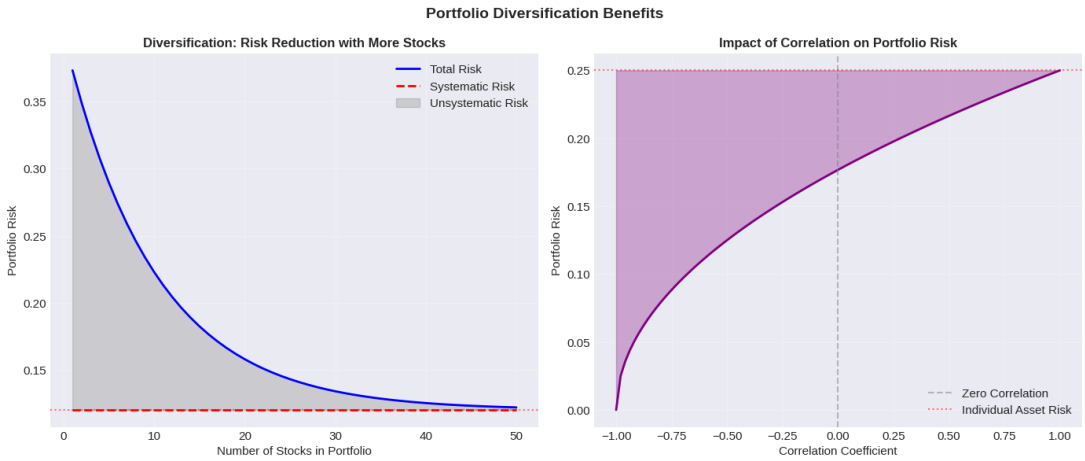


Figure 9: Portfolio Diversification Benefits - Risk reduction with more stocks and impact of correlation on portfolio risk

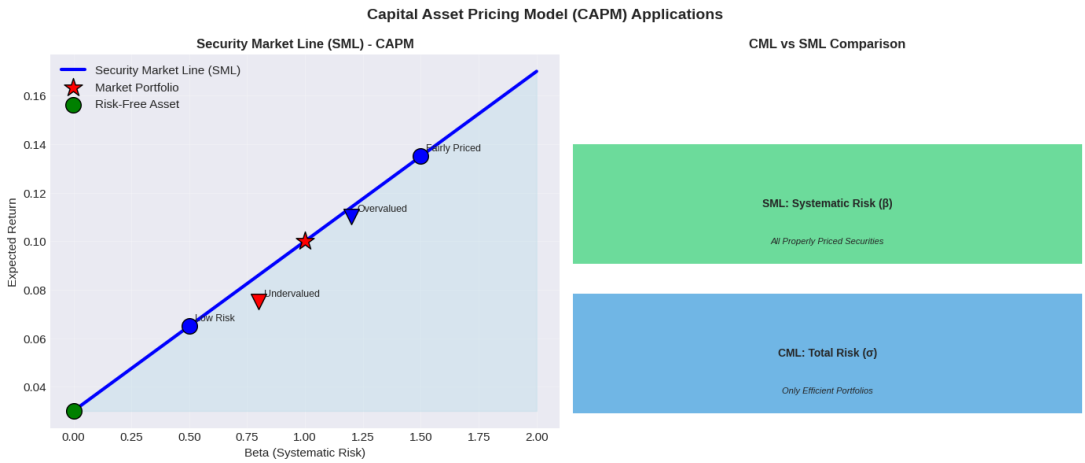


Figure 10: Capital Asset Pricing Model (CAPM) Applications - Security Market Line and CML vs SML comparison

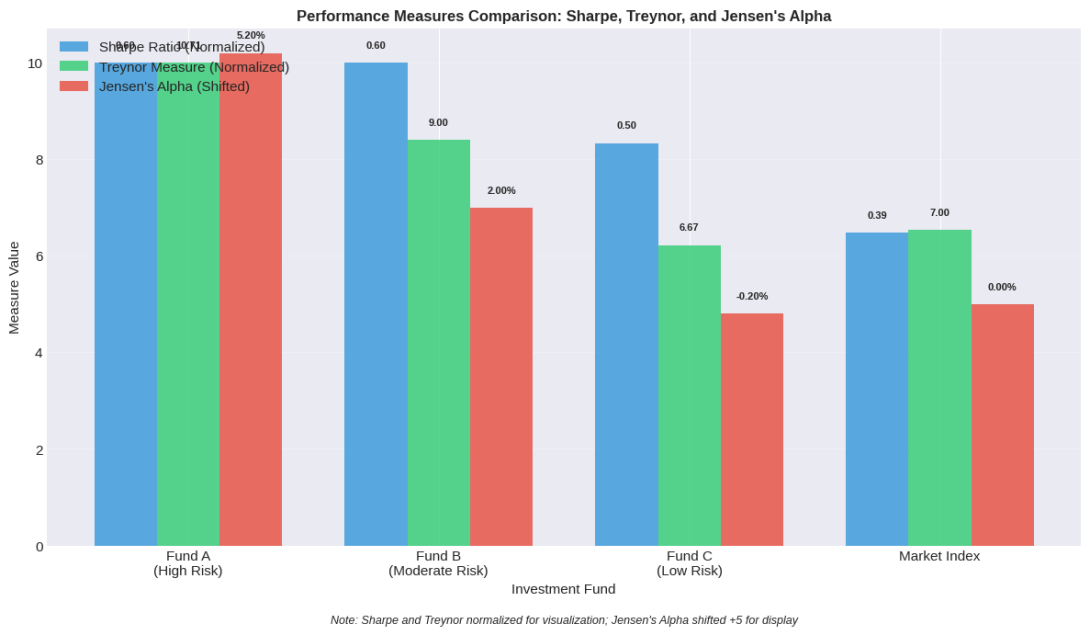


Figure 11: Performance Measures Comparison - Sharpe Ratio, Treynor Measure, and Jensen's Alpha

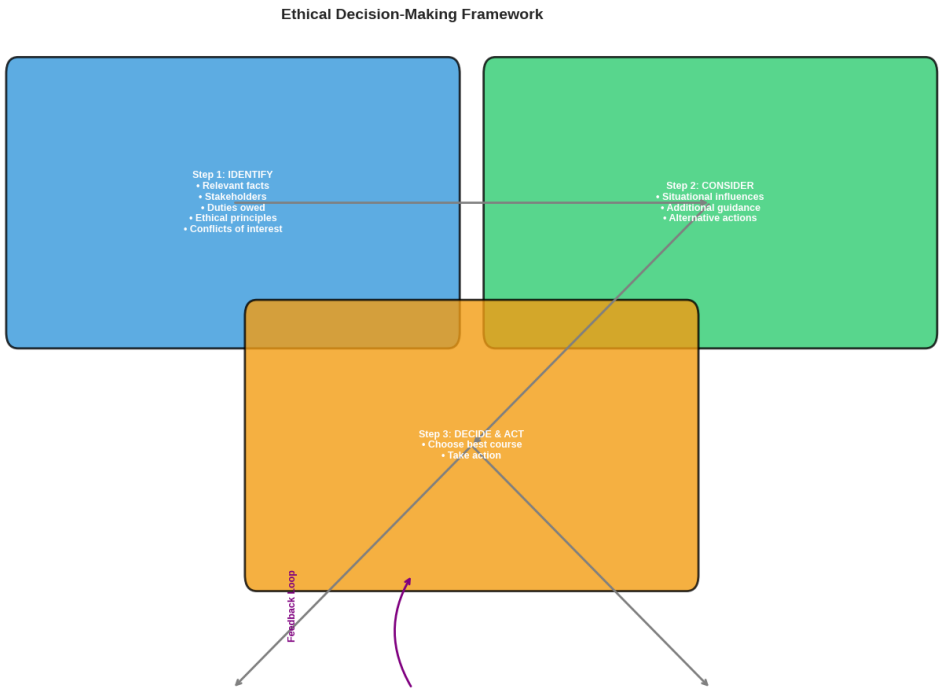


Figure 12: Ethical Decision-Making Framework - 4-step process for ethical decision making

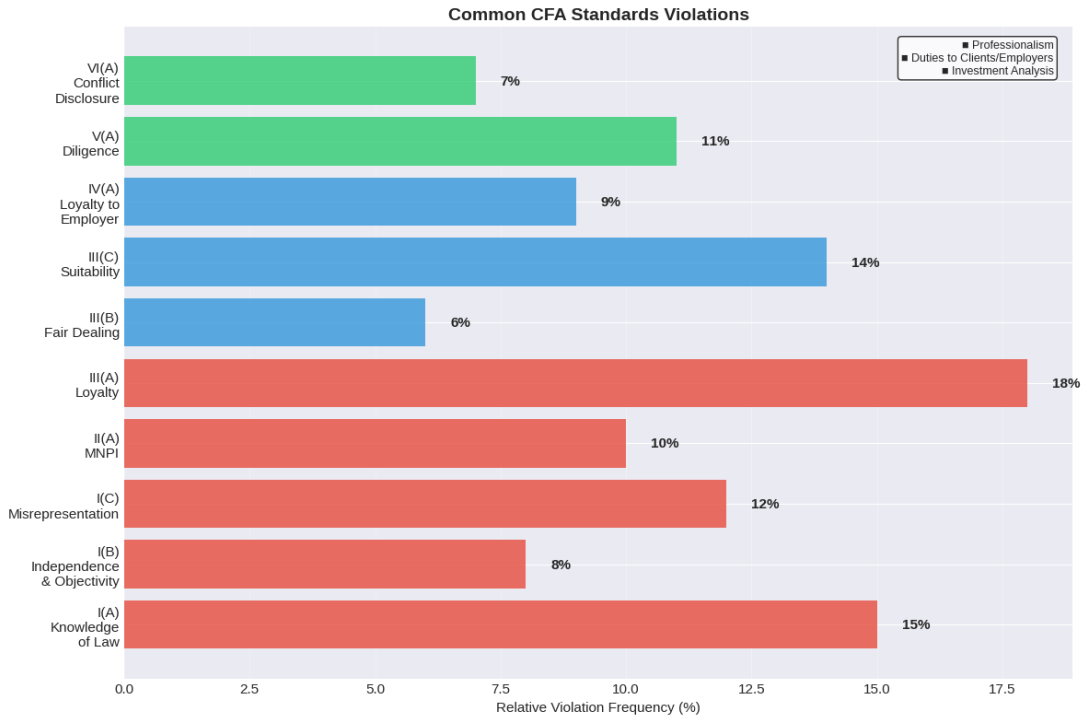


Figure 13: Common CFA Standards of Professional Conduct Violations

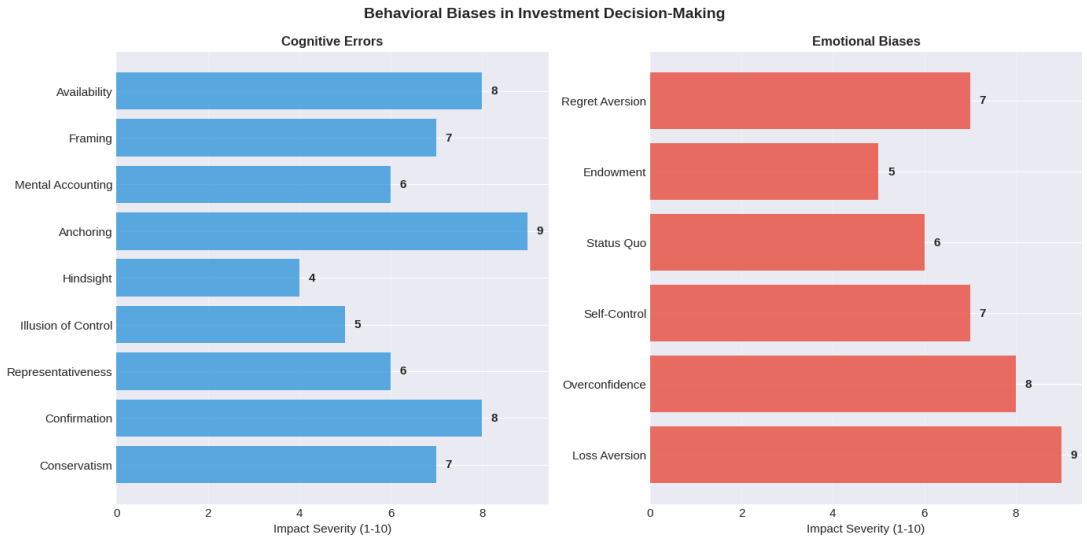


Figure 14: Behavioral Biases - Cognitive Errors vs Emotional Biases in Investment Decision-Making